Inverse of a matrix

Suppose A is an $n \times n$ matrix. We say that A is invertible if there exists an $n \times n$ matrix C such that

$$AC = CA = \mathrm{I}_n$$

The matrix C is called the inverse of A, and is usually denoted by A^{-1}.

- If a matrix is invertible, then it has a unique inverse.
- If A and B are $n \times n$ and $AB = \mathrm{I}_n$, then automatically we also have $BA = \mathrm{I}_n$.

Theorem (2×2 matrices)

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then :

- If $ad - bc = 0$ then A is **not** invertible.
- If $ad - bc \neq 0$ then A is invertible and we have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Finding A^{-1} for general A

Recall that I_n is the $n \times n$ identity matrix, i.e.,

$$\mathrm{I}_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Algorithm for finding A^{-1}

Suppose A is an $n \times n$ matrix.

Row reduce the super augmented matrix $[\, A \mid \mathrm{I}_n \,]$ to obtain its R.E.F.

- If we obtain a matrix of the form $[\, \mathrm{I}_n \mid B \,]$, then A is invertible and $B = A^{-1}$.
- Otherwise, A is not invertible.

Example

Suppose $A = \begin{bmatrix} 1 & 0 & -2 \\ 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix}$ Is A invertible?

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & -3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightsquigarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right]$$

$$\begin{matrix} R_1 \to R_1 + 2R_3 \\ R_2 \to R_2 - 2R_3 \end{matrix} \xrightarrow{\qquad} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -2 & 0 \\ 0 & 1 & 0 & -2 & 2 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right] \qquad \textit{R.E.F.}$$

Therefore A is invertible and $A^{-1} = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix}$

- It is always a good idea to check the answer:

$$AA^{-1} = \begin{bmatrix} 1 & 0 & -2 \\ 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & -2 & 0 \\ -2 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \checkmark$$

(We do not need to check $A^{-1}A = \mathrm{I}_3$.)

Example

Is $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix}$ invertible? If it is, then find its inverse.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -2 & 3 & -2 & 1 & 0 \\ 0 & -2 & 3 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -2 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right] \xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -2 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right]$$

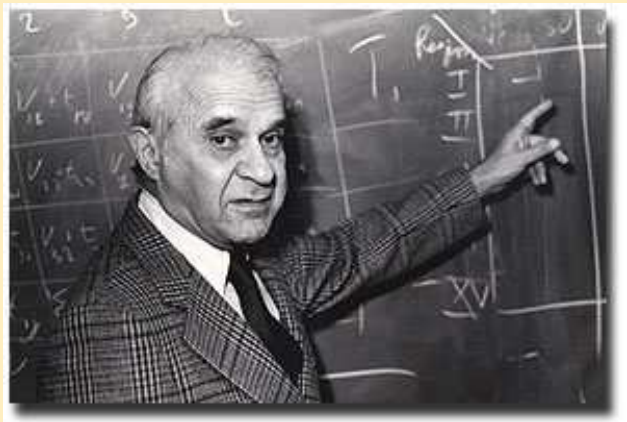
$$\xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & -1 & 1 \\ 0 & -2 & 3 & 0 & 3 & -2 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & -1 & 1 \\ 0 & 1 & -\frac{3}{2} & 0 & -\frac{3}{2} & 1 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{3}{2} & 0 & -\frac{3}{2} & 1 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right] \quad \textit{R.E.F.}$$

It follows that A is not invertible.

Wassily Leontief (1905-1999)

- Born in Germany, Raised in Russia.
- Professor at Harvard University.
- Nobel Laureate in Economics – 1973 for development of Input-Output models.



Input-Output model : A model for analyzing the interrelation between production of economic sectors.

Features :

1. Formulates the relation between economic sectors as a system of linear equations.
2. Can be solved using computers.
3. Its variations have had practical use for about fifty years.

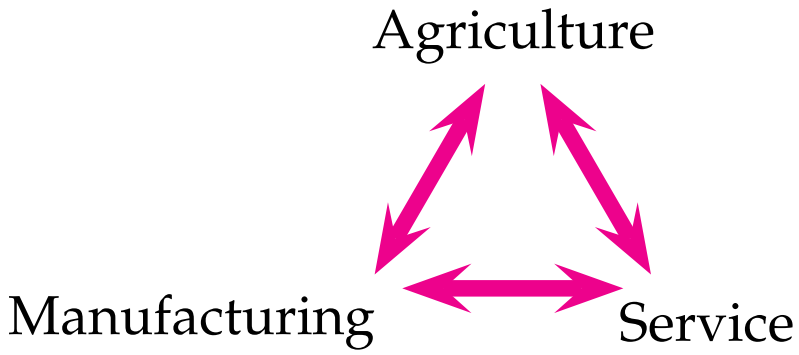
- Consider an economy with several sectors :

Sector 1, Sector 2,..., Sector n.

1. Each sector can produce goods (and/or services).
2. The product of every sector is evaluated by a common unit price (e.g., millions of dollars).
3. The production of each sector requires consumption of products of the other sectors.
4. The goal of the economy is to meet consumers' demands, i.e., to provide a final output (called the final demand) for the open sector.

   Open sector : The consumers' sector, i.e. a sector that does not produce goods.

Example. Consider an economy with three sectors : Agriculture, Manufacturing, Service.



Unit consumption vector

For each sector, there is a unit consumption vector which lists the *input* from other sectors required for an output of one unit of goods/services.

Example

Suppose the unit consumption vector of Manufacturing is $\begin{bmatrix} .2 \\ .3 \\ .4 \end{bmatrix}$.

This means that to produce one unit (e.g., one million dollar) worth of manufacturing products, the manufacturing sector needs to consume

1. .2 units from Agriculture products.
2. .3 units from Manufacturing products.
3. .4 units from Service products.

Example

Suppose that the unit consumption vector of Manufacturing is $\begin{bmatrix} .2 \\ .3 \\ .4 \end{bmatrix}$.

What amounts will be consumed by the manufacturing sector if it decides to produce 50 units?

Solution. $50 \begin{bmatrix} .2 \\ .3 \\ .4 \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \\ 20 \end{bmatrix}$ i.e.,

1. 10 units from Agriculture.
2. 15 units from Manufacturing.
3. 20 units from Service.

The interrelation of various sectors can be represented by a table :

	Inputs consumed per unit of Output		
Purchased from	**Agriculture**	**Manufacturing**	**Service**
Agriculture	.6	.2	.2
Manufacturing	.1	.3	.1
Service	0	.4	.6
	$\uparrow$ $\mathbf{c}_1$	$\uparrow$ $\mathbf{c}_2$	$\uparrow$ $\mathbf{c}_3$

Example

What amounts will be consumed by the Service sector if it decides to produce 25 units?

Solution. $25\mathbf{c}_3 = 25\begin{bmatrix}.2\\.1\\.6\end{bmatrix} = \begin{bmatrix}5\\2.5\\15\end{bmatrix} \implies \begin{cases} \bullet \text{ 5 units from Agriculture} \\ \bullet \text{ 2.5 units from Manufacturing} \\ \bullet \text{ 15 units from Service} \end{cases}$

	Inputs consumed per unit of Output		
Purchased from	**Agriculture**	**Manufacturing**	**Service**
Agriculture	.6	.2	.2
Manufacturing	.1	.3	.1
Service	0	.4	.6
	↑ $\mathbf{c}_1$	↑ $\mathbf{c}_2$	↑ $\mathbf{c}_3$

Example

What amounts will be consumed if the Manufacturing decides to produce 10 units and the Service sector decides to produce 15 units?

Solution.

$$10\mathbf{c}_2 + 15\mathbf{c}_3 = 10\begin{bmatrix}.2\\.3\\.4\end{bmatrix} + 15\begin{bmatrix}.2\\.1\\.6\end{bmatrix} = \begin{bmatrix}5\\4.5\\13\end{bmatrix} \Longrightarrow \begin{cases}\text{5 units from Agriculture}\\\text{4.5 units from Manufacturing}\\\text{13 units from Service}\end{cases}$$

	Inputs consumed per unit of Output		
Purchased from	**Agriculture**	**Manufacturing**	**Service**
Agriculture	.6	.2	.2
Manufacturing	.1	.3	.1
Service	0	.4	.6
	↑ $\mathbf{c}_1$	↑ $\mathbf{c}_2$	↑ $\mathbf{c}_3$

More generally, if

- Agriculture decides to produce x_1 units,
- Manufacturing decides to produce x_2 units, and
- Service decides to produce x_3 units,

then the total intermediate demand is equal to

$$x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + x_3\mathbf{c}_3.$$

	Inputs consumed per unit of Output		
Purchased from	**Agriculture**	**Manufacturing**	**Service**
Agriculture	.6	.2	.2
Manufacturing	.1	.3	.1
Service	0	.4	.6
	↑ $\mathbf{c}_1$	↑ $\mathbf{c}_2$	↑ $\mathbf{c}_3$

- Another way to write the total intermediate demand is the following :

$$x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + x_3\mathbf{c}_3 = [\ \mathbf{c}_1\ \ \mathbf{c}_2\ \ \mathbf{c}_3\ ] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = C\mathbf{x}$$

where

$$C = [\ \mathbf{c}_1\ \ \mathbf{c}_2\ \ \mathbf{c}_3\ ] \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- The matrix C is called the consumption matrix.

We want to answer the following question :

Question. how many units of output should each sector produce to meet the consumer's demand, i.e., to obtain a *final demand* $\mathbf{d}$?

- Since there are three sectors in our example, the final demand $\mathbf{d}$ should be represented by a vector in $\mathbb{R}^3$. (If there are n sectors, the final demand $\mathbf{d}$ would be a vector in $\mathbb{R}^n$.)

Leontief's Input-Output Equation

$$\underbrace{\text{amount produced}}_{\mathbf{x}} - \underbrace{\text{intermediate demand}}_{C\mathbf{x}} = \underbrace{\text{final demand}}_{\mathbf{d}}$$

- Mathematically, we want to solve the following equation for $\mathbf{x}$:

$$\mathbf{x} - C\mathbf{x} = \mathbf{d} \quad \text{or, equivalently,} \quad (\mathrm{I} - C)\mathbf{x} = \mathbf{d}$$

	Inputs consumed per unit of Output		
Purchased from	**Agriculture**	**Manufacturing**	**Service**
Agriculture	.6	.2	.2
Manufacturing	.1	.3	.1
Service	0	.4	.6
	↑ $\mathbf{c}_1$	↑ $\mathbf{c}_2$	↑ $\mathbf{c}_3$

Example

Problem. Find the production levels that meet a final demand of 24 units from Agriculture, 10 units from Manufacturing, and 35 units from Service.

Solution. $C = \begin{bmatrix} .6 & .2 & .2 \\ .1 & .3 & .1 \\ 0 & .4 & .6 \end{bmatrix} \implies I - C = \begin{bmatrix} .4 & -.2 & -.2 \\ -.1 & .7 & -.1 \\ 0 & -.4 & .4 \end{bmatrix}$

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. We want to solve $(I - C)\mathbf{x} = \mathbf{d}$, i.e., $\begin{bmatrix} .4 & -.2 & -.2 \\ -.1 & .7 & -.1 \\ 0 & -.4 & .4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 24 \\ 10 \\ 35 \end{bmatrix}$

$$\implies \mathbf{x} = \begin{bmatrix} .4 & -.2 & -.2 \\ -.1 & .7 & -.1 \\ 0 & -.4 & .4 \end{bmatrix}^{-1} \begin{bmatrix} 24 \\ 10 \\ 35 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 2 \\ .5 & 2 & .75 \\ .5 & 2 & 3.25 \end{bmatrix} \begin{bmatrix} 24 \\ 10 \\ 35 \end{bmatrix} = \begin{bmatrix} 162 \\ 58.25 \\ 145.75 \end{bmatrix}$$

How to solve $\begin{bmatrix} .4 & -.2 & -.2 \\ -.1 & .7 & -.1 \\ 0 & -.4 & .4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 24 \\ 10 \\ 35 \end{bmatrix}$ efficiently?

We should find the R.E.F. of the augmented matrix :

$$\left[\begin{array}{ccc|c} .4 & -.2 & -.2 & 24 \\ -.1 & .7 & -.1 & 10 \\ 0 & -.4 & .4 & 35 \end{array}\right] \xrightarrow[R_3 \to 10R_3]{R_1 \to 10R_1 \\ R_2 \to 10R_2} \left[\begin{array}{ccc|c} 4 & -2 & -2 & 240 \\ -1 & 7 & -1 & 100 \\ 0 & -4 & 4 & 350 \end{array}\right]$$

Finally we have :

$$\left[\begin{array}{ccc|c} 4 & -2 & -2 & 240 \\ -1 & 7 & -1 & 100 \\ 0 & -4 & 4 & 350 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 162 \\ 0 & 1 & 0 & 58.25 \\ 0 & 0 & 1 & 145.75 \end{array}\right]$$

$$\Longrightarrow \mathbf{x} = \begin{bmatrix} 162 \\ 58.25 \\ 145.75 \end{bmatrix}$$

- Agriculture must produce 162 units,
- Manufacturing must produce 58.25 units, and
- Service must produce 145.75 units.

Input-Output Equation (cont.)

The equation

$$\mathbf{x} - C\mathbf{x} = \mathbf{d}$$

can be written as

$$\mathrm{I}\,\mathbf{x} - C\mathbf{x} = \mathbf{d} \quad \Longrightarrow \quad (\mathrm{I} - C)\mathbf{x} = \mathbf{d}$$

If $\mathrm{I} - C$ is invertible, we can write

$$\mathbf{x} = (\mathrm{I} - C)^{-1}\mathbf{d}$$

Admissible Solution

Recall that we want to solve the equation

$$(\mathrm{I} - C)\mathbf{x} = \mathbf{d}$$

- We know that production from a sector cannot be negative, i.e. entries of $\mathbf{x}$ must be nonnegative numbers.

Theorem

Let C be the consumption matrix for an economy, and let $\mathbf{d}$ be the final demand. If

- the entries of C and $\mathbf{d}$ are nonnegative, and
- the sum of the entries of each column of C is less than 1,

then $(\mathrm{I} - C)$ is invertible and the production vector $\mathbf{x} = (\mathrm{I} - C)^{-1}\mathbf{d}$ has nonnegative entries.

Example:

Example

The economy of the small European country of Liberlandt has three major sectors : Agriculture, Chemicals, Manufacturing .
Suppose that :

- Each unit produced by Agriculture requires **.6** units from Agriculture, **.2** units from Chemicals and **.1** units from Manufacturing.
- Each unit produced by Chemicals requires **.2** units from Agriculture, **.6** units from Chemicals and **.1** units from Manufacturing.
- Each unit produced by Manufacturing requires **.1** units from Agriculture, **.1** units from Chemicals and **.6** units from Manufacturing.

Question.

1. What intermediate demands are created if Chemicals decides to produce 16 units?
2. Determine the production levels needed to meet the following final demands:
   - 30 units from Chemicals and no demand from other sectors.
   - 24 units from Agriculture, 18 units from Chemicals, and 60 units from manufacturing.

Solution of the Example:

Example

The economy of the small European country of Liberlandt has three major sectors : Agriculture, Chemicals, Manufacturing .
Suppose that :

- Each unit produced by Agriculture requires **.6** units from Agriculture, **.2** units from Chemicals and **.1** units from Manufacturing.
- Each unit produced by Chemicals requires **.2** units from Agriculture, **.6** units from Chemicals and **.1** units from Manufacturing.
- Each unit produced by Manufacturing requires **.1** units from Agriculture, **.1** units from Chemicals and **.6** units from Manufacturing.

	Inputs consumed per unit of Output		
Purchased from	**Agriculture**	**Chemicals**	**Manufacturing**
Agriculture	.6	.2	.1
Chemicals	.2	.6	.1
Manufacturing	.1	.1	.6
	$\uparrow$	$\uparrow$	$\uparrow$
	$\mathbf{c}_1$	$\mathbf{c}_2$	$\mathbf{c}_3$

Solution of the Example:

	Inputs consumed per unit of Output		
Purchased from	**Agriculture**	**Chemicals**	**Manufacturing**
Agriculture	.6	.2	.1
Chemicals	.2	.6	.1
Manufacturing	.1	.1	.6
	$\uparrow$ $\mathbf{c}_1$	$\uparrow$ $\mathbf{c}_2$	$\uparrow$ $\mathbf{c}_3$

Example

1. What intermediate demands are created if Chemicals decides to produce 16 units?

Solution. $16\mathbf{c}_2 = 16\begin{bmatrix} .2 \\ .6 \\ .1 \end{bmatrix} = \begin{bmatrix} 3.2 \\ 9.6 \\ 1.6 \end{bmatrix}$

Solution of the example (cont.)

Example

2. Determine the production levels needed to meet the following final demands:
   - 30 units from Chemicals and no demand from other sectors.
   - 24 units from Agriculture, 18 units from Chemicals, and 60 units from manufacturing.

Solution.

$$C = \begin{bmatrix} .6 & .2 & .1 \\ .2 & .6 & .1 \\ .1 & .1 & .6 \end{bmatrix}$$

$$I - C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} .6 & .2 & .1 \\ .2 & .6 & .1 \\ .1 & .1 & .6 \end{bmatrix} = \begin{bmatrix} .4 & -.2 & -.1 \\ -.2 & .4 & -.1 \\ -.1 & -.1 & .4 \end{bmatrix}$$

Solution of the example (cont.)

Example

2. Determine the production levels needed to meet the following final demands:
   - 30 units from Chemicals and no demand from other sectors.
   - 24 units from Agriculture, 18 units from Chemicals, and 60 units from manufacturing.

To calculate $(\mathrm{I} - C)^{-1}$, we need to calculate the R.E.F. of $\left[\begin{array}{c|c} \mathrm{I} - C & \mathrm{I} \end{array}\right]$.

$$\left[\begin{array}{c|c} \mathrm{I} - C & \mathrm{I} \end{array}\right] = \left[\begin{array}{ccc|ccc} .4 & -.2 & -.1 & 1 & 0 & 0 \\ -.2 & .4 & -.1 & 0 & 1 & 0 \\ -.1 & -.1 & .4 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} .4 & -.2 & -.1 & 1 & 0 & 0 \\ -.2 & .4 & -.1 & 0 & 1 & 0 \\ -.1 & -.1 & .4 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{25}{6} & \frac{5}{2} & \frac{5}{3} \\ 0 & 1 & 0 & \frac{5}{2} & \frac{25}{6} & \frac{5}{3} \\ 0 & 0 & 1 & \frac{5}{3} & \frac{5}{3} & \frac{10}{3} \end{array}\right]$$

It follows that : $(\mathrm{I} - C)^{-1} = \left[\begin{array}{ccc} \frac{25}{6} & \frac{5}{2} & \frac{5}{3} \\ \frac{5}{2} & \frac{25}{6} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} & \frac{10}{3} \end{array}\right]$

Solution of the example (cont.)

Example

2. Determine the production levels needed to meet the following final demands :

   - 30 units from Chemicals and no demand from other sectors.

     Solution.

     $$\mathbf{d} = \begin{bmatrix} 0 \\ 30 \\ 0 \end{bmatrix} \Longrightarrow \mathbf{x} = (\mathrm{I} - C)^{-1}\mathbf{d} = \begin{bmatrix} \frac{25}{6} & \frac{5}{2} & \frac{5}{3} \\ \frac{5}{2} & \frac{25}{6} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} & \frac{10}{3} \end{bmatrix} \begin{bmatrix} 0 \\ 30 \\ 0 \end{bmatrix} = \begin{bmatrix} 75 \\ 125 \\ 50 \end{bmatrix}$$

   - 24 units from Agriculture, 18 units from Chemicals, and 60 units from manufacturing.

     Solution.

     $$\mathbf{d} = \begin{bmatrix} 24 \\ 18 \\ 60 \end{bmatrix} \quad \Longrightarrow \quad \mathbf{x} = (\mathrm{I} - C)^{-1}\mathbf{d} = \begin{bmatrix} \frac{25}{6} & \frac{5}{2} & \frac{5}{3} \\ \frac{5}{2} & \frac{25}{6} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} & \frac{10}{3} \end{bmatrix} \begin{bmatrix} 24 \\ 18 \\ 60 \end{bmatrix} = \begin{bmatrix} 245 \\ 235 \\ 270 \end{bmatrix}$$